Decision Analytics Journal 9 (2023) 100364

Contents lists available at ScienceDirect

Decision Analytics Journal

journal homepage: www.elsevier.com/locate/dajour

# A global cyber-threat intelligence system with artificial intelligence and convolutional neural network

Fahim Sufi

*Monash University, Melbourne, VIC 3004, Australia*

ARTICLE INFO

*Keywords:*
Cyber threat intelligence
Cyber threat prediction
Decision support system
Cyber anomaly detection
Cyber attack dashboard
Social-media analysis

ABSTRACT

Global cyber-attacks significantly impact the economy, society, organizations, and individuals. Existing research on cyber-attacks lacks in demonstrating Artificial Intelligence (AI) based analytical solutions for providing country-wide cyber threat intelligence. Cyber strategists at a national level require AI-based decision support systems for deciding a country's cyber posture or preparedness. This paper proposes an AI-based solution that autonomously collects multidimensional cyber-attack data on social media posts on cyber-related outcry. The proposed system provides critical analytical capability in the cyber-threat spectrum and uses sophisticated AI-based algorithms for anomaly detection, prediction, sentiment analysis, location detection, translation, etc. The proposed system was deployed from 11 October 2022 to 31 October 2022. During these 21 days, the system autonomously collected 30,203 records on cyber threats covering multiple dimensions of cyber-threat. These dimensions included country-wide daily cyber-attack records by ransomware, exploits, web threats, spam, malicious mail, network attacks, local infections, and on-demand scan. Moreover, the system performed AI-based acquisition and analysis of 3789 cyber-related tweets from 3402 tweet users in 37 different languages. The system also autonomously translated 893 non-English tweets. The proposed system is the first solution that uses Convolutional Neural Network (CNN) based anomaly detection to detect abnormalities in cyber-threat spectrum worldwide and predict cyber-attacks automatically. The proposed system was demonstrated to provide evidence-based decisions on global cyber threats in multiple platforms, including iOS, Android, and Windows.

## 1. Introduction

Worldwide Cyber-attacks had costed almost 1 trillion USD to global economy in 2020 [1]. Cybercrimes are projected to cost as much as 10.5 trillion USD annually by 2025 [2]. Larger economies like China, Brazil, United States, India, Mexico, France, Australia and even United Arab Emirates suffer from billions of dollars in consumer loss through cybercrimes [3]. For example, China's consumer loss through cyber-crime was 66.3 billion USD in 2017 [3]. Other than creating economic losses, Cyber-attacks has detrimental effect at social and psychological level touching the lives of human being [4]. For example, very recently one of Australia's telecom giant, Optus have been exposed to data breach as a result of cyber-attack, which has created massive stress and anger on consumers [5,6]. Because of this cyber-attack, information of Optus customers (e.g., name, date of birth, email addresses, driver's license, Medicare card, passport numbers etc.) have been exposed causing distress among 2.1 million consumers [6,7]. Most recently, sensitive patient information that include medical diagnosis & procedure were stolen by cyber criminals from Australia's number 1 health insurer [8]. Hence, cyber-crime is one of the most critical concerns of modern nations, states, organizations, and individuals.

To reduce the impact of cyber-crimes two critical mandates are aggregation of cyber related data and sophisticated analytical algorithms to identify and deter threats. The requirements of cyber data have been portrayed in recent research works [1,9]. On the other hand, the requirement of Artificial Intelligence (AI) based sophisticated analytical algorithms have been demonstrated in [10–19]. While AI can be used to conduct sophisticated cyber-attacks [10,11], AI can also be used in detecting cyber-attack as demonstrated in [11–19]. However, none of these existing literatures provide country-level historical cyber statistics, country level cyber prediction, and anomaly detections with AI. Therefore, the proposed cyber intelligence dashboard contributes towards the following Critical Requirements (CR) to cyber strategists for making evidence-based decisions on national cyber posture:

**CR 1:** Analyze threat spectrum against different dimensions of cyber-attacks for a particular country and compare with other countries in the globe

**CR 2:** Analyze changes of threat spectrum by time for any countries (e.g., daily, weekly, monthly change)

**CR 3:** Predict cyber-attacks for any countries in the globe by analyzing past cyber-attack data

*E-mail address:* fahim.sufi@monash.edu.

https://doi.org/10.1016/j.dajour.2023.100364
Received 31 October 2022; Received in revised form 31 October 2023; Accepted 10 November 2023
Available online 14 November 2023
2772-6622/© 2023 The Author(s). Published by Elsevier Inc. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).

**Fig. 1.** Aggregation of multidimensional cyber-attack data and cyber related social media data.

**CR 4:** Detect Anomalies in cyber-threat spectrums for any countries in the globe by analyzing past cyber-attack data

**CR 5:** Analyze the view of social media users on various cyber issues over time

**CR 6:** Obtain the above analytical requirements (i.e., CR 1 to CR 5) in any platforms (i.e., Windows, Android, iOS) on any devices (e.g., mobile, tablet, desktop)

In this paper, we propose an innovative solution that successfully meets the analytical requirements highlighted within CR 1 to CR 6, by aggregating cyber related data from multiple sources and using AI-based services and algorithms. As seen from Fig. 1, multidimensional Cyber-attack data were obtained from leading Anti-virus software vendor (e.g., Kaspersky [20]) and Cyber related social media data was obtained from Twitter. As seen from Fig. 1, the multidimensional cyber-attack statistics included Ransomware [21], Exploits [22], Web threats [23], Spam [24], Malicious mail [25], Network attacks [26], Local Infections [27], and On-demand scan [28] for all the countries in the globe.

## 2. Background literature

Existing literatures in Cyber-Threat intelligence mainly focused on an organizational perspective, where cyber-attack impacted an individual organization for financial or political benefit by the perpetrator [12–19]. While these perspective serves the concerns of an individual organizational entity, this does not cater the need of political heads of states having a wholistic view on their nation. National level concern on cyber-attack requires country-wide cyber threat intelligence that includes attack spectrum covering an individual country along with other countries in the globe for benchmarking purpose. Using this wholistic view of all the cyber-attack statistics for a country in comparison with other countries allows the political decision maker to determine their posture on cyber defense. For example, if Australia is facing unusually higher number of cyber-attacks compared to other nations like New Zealand, United Arab Emirates, Malaysia, then Australia should place their Cyber defense posture at a higher level. Country-wide cyber threat intelligence is critical for a nation's Cyber Preparedness.

Existing research also suffers from not utilizing AI-based fully automated data acquisition techniques like Web Scraping, Entity Detection for Location Intelligence, Sentiment Analysis as their data requirements were primarily focused on intrusion detections within the network. As a result, most of the existing studies as demonstrated in [12,13,16,17,29], were only obtaining Network Traffic Data. Study in [18] used simulated sensor data on cyber physical system. On the other hand, researcher in [30] used survey data collected from 294 participants. None of these existing research works, obtained cyber threat data from multiple sources like Cyber-attack statistics for all the countries from Anti-Virus Software vendors or Cyber related outcry from live social media posts. Social media data on geopolitical relevance requires AI-based data acquisition and pre-processing as demonstrated in our recent papers [31–38]. As social media contents were fetched, our novel methodology demonstrated in [31–38], applied AI-based Named Entity Recognition (NER) for extracting geospatial intelligence, sentiment analysis for extracting subjective relevance of the content, dynamic translation to understand social media contents in 110 different languages.

Existing literature demonstrates that there have been a growing interest using AI-based Anomaly detection and other deep learning methodologies for detecting cyber-attack [11–19]. However, within these studies Anomaly detection has been used on Network Traffic data for detecting intrusions. Anomaly detection algorithms have never been reported to be used on country-wide cyber-attack statistics.

Finally, almost none of the existing literature on cyber-attack provided interactive dashboards that updates itself automatically for providing evidence-based decisions on cyber posture for strategic leaders of a nation. Since a strategic decision maker needs to make instant decisions, the dashboards are required to be available in a wide range of platforms that includes mobile, tablet or traditional desktops. Existing research work on Cyber-attack depicted in [11–19] were not available a wide range of platforms like Windows, iOS, or Android. It should be mentioned that the requirement of strategic decision makers to make instant decision on a wide range of platforms like mobile phone running iOS or Android have been demonstrated in [31–37,39–43]. Table 1, summarizes the bottleneck of existing studies and clearly demonstrates how the solution presented in this paper aims to address these shortcomings.

**Table 1**
Literature review of existing studies in Cyber threat analysis with supported features.

| Reference | 1. Country-Wide cyber threat intelligence | 2. Automated AI-Based data acquisition | 3. Use of multi-source data | 4. AI-based deep learning/ anomaly detection/ prediction | 5. Self-updating dashboard | 6. Interactive dashboard for decision making | 7. Dashboard availability in multiple platforms |
|---|---|---|---|---|---|---|---|
| [12] | No | No | No. Network Traffic Data | Yes | No | No | No |
| [13] | No | No | No. Network Traffic Data | Yes | No | No | No |
| [29] | No | No | No. Network Traffic Data | Yes | No | No | No |
| [16] | No | No | No | Yes | Yes | No | No |
| [17] | No | No | No | Yes | No | No | No |
| [18] | No | No | No. Simulated Sensor Data | Yes | No | No | No |
| [19] | No | No | No. Network Traffic Data | No | Yes | Yes | No |
| [30] | No | No | No. Survey Data | No. Only Statistical Methods | No | No | No |
| **Proposed** | Yes | Yes | Yes | Yes | Yes | Yes | Yes |

**Fig. 2.** Architecture of acquiring Multidimensional Cyber Attack data and Cyber relevant social media data for rich data visualization and analytics.

As seen from Table 1, the proposed approach presented in this paper automatically acquires multidimensional data on cyber-attack covering dimensions such as Spam, Ransomware, Web Threats, Network Attacks, Malicious Mails, Local Infections, Exploits, and On Demand Scans. Moreover, the presented approach obtains cyber related posts from live social media from Twitter for having a comprehensive understanding of cyber threats that social media users are talking about. Furthermore, this paper applies AI-based deep learning techniques (i.e., Anomaly Detection) on global Country-Wide attack data to automatically identify abnormally high or low level of attacks on a nation. Finally, the presented solution is deployed into multiple platforms like desktop, tablet, and mobile in a wide range of environments like iOS, Android, Windows. Hence, strategic decision-makers can make instant evidence-based decisions right from their device of choice. The fully interactive dashboard updates (i.e., refreshes itself) providing the most up-to-date information supporting data driven decision support required by today's strategic leaders [44].

## 3. Material & methods

Multidimensional Global Cyber Attack data is acquired using a combination of Desktop Flow and Cloud Flow in Power Automate [45]. The Desktop flow makes use of unattended Power Automate Desktop flow for automating a series of tasks. This automation opens up browser, goes to 8 different links [21–28] downloads 8 different attack statistics files, and saves these files (as Microsoft Excel .xlsx) into designated One Drive for business folder. With every new file creation, a trigger initiates a separate Microsoft Power Automate cloud flow that processes these attack statistics and stores them in Microsoft Dataverse [46]. Thus, by using Desktop flow and Cloud Flow, the presented system automatically and anonymously updates itself on a daily schedule with daily cyber statistics provided by [21–28]. It should be noted that the Kaspersky's sites in [21–28] does not provide access to historical data. Every day, the daily cyber statistics are replaced with updated statistics. Hence, the presented system builds historical cyber statistics on a daily basis (i.e., not in real-time). However, the presented system would support real-time updates (using the same technology stack of Microsoft Power Automate), when the cyber-attack statistics are provided in real-time (by the data sources).

On the other hand, social media posts from Twitter are extracted using Microsoft Power Automate flow. During this extractions AI-based Sentiment Analysis, NER, Translation is performed as depicted in our most recent research [31–38,42]. Finally, these records are stored in Microsoft SQL Server Database.

Microsoft Power BI [47] obtains both these data sources (i.e., multidimensional cyber threat data of Kaspersky and cyber related social media posts from Twitter) to provide data analytics and data visualizations for the strategic decision makers in iOS, Windows, and Android platforms. Fig. 2 depicts this overall process. Furthermore, Table 2 provide details into various technology components used within the proposed system along with justifications. Table 2 refers to individual features that was provided within Table 1 and maps those features to technology components.

As it appears from Fig. 2 and Table 1, AI-based anomaly detection is the key technology empowering the decision analytics. The anomaly detection adds additional feature to line charts by automatically detecting anomalies within time-series data. It also provides Natural Language Processing (NLP) based [48] explanations for the anomalies facilitating root-cause analysis. In our most recent study, we have used AI-based anomaly detection for identifying abnormal cases of